

Algorithm 4 Evolved mutation: duplicate and couple a reusable motif

Input Expression tree t ; available binary operators $\mathcal{O}_2$; maximum node count L ; number of features d .

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1  $\mathcal{O}_a \leftarrow \{+, \times\} \cap \mathcal{O}_2$ 
2  $\delta \leftarrow -$  if available, otherwise  $+$  if available, otherwise  $\emptyset$ 
3  $r_{\text{ok}} \leftarrow [/\in \mathcal{O}_2] \wedge [\delta \neq \emptyset] \wedge [L - |t| - 3 \geq 1]$ 
4  $r \leftarrow r_{\text{ok}} \wedge (\mathcal{O}_a = \emptyset \vee \text{Unif}(0, 1) < 0.35)$ 
5 If  $\neg r$  and  $\mathcal{O}_a = \emptyset$ , return  $t$ 
6  $b \leftarrow L - |t| - (3 \text{ if } r \text{ else } 1)$  # Node budget for the donor
7  $\mathcal{D} \leftarrow \{u \text{ subtree of } t : |u| \leq b, u \text{ contains a variable}\}$ 
8 If  $\mathcal{D} = \emptyset$ , return  $t$ 
9  $\mathcal{D}_+ \leftarrow \{u \in \mathcal{D} : u \text{ is not a leaf}\}$ 
10  $v \leftarrow \text{COPY}(\text{Unif}(\mathcal{D}_+ \text{ if nonempty, else } \mathcal{D}))$ 
11  $u \leftarrow$  a uniformly sampled target node of  $t$ 
12 if  $d > 1$  and  $\text{Unif}(0, 1) < 1/2$  then
13 |  $h \leftarrow \text{Unif}\{1, \dots, d-1\}$ 
14 | In  $v$ , replace every feature  $x_j$  by  $x_{1+((j-1+h) \bmod d)}$ 
15 if  $r$  then
16 | Replace  $u$  by  $\text{COPY}(u)/(1 \delta v)$  # Rational correction
17 else
18 |  $\circ \leftarrow \text{Unif}(\mathcal{O}_a)$ ; replace  $u$  by  $\text{COPY}(u) \circ v$ 
19 return  $t$ 

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The donor is copied before editing the target, so the two may overlap in the original tree. Compound, variable-bearing donors are preferred; the cyclic shift reuses their structure across features. The 0.35 rational probability applies when rational coupling is feasible and an accumulator exists. If rational coupling is the only option, it is always chosen. The outer PySR mutation machinery still checks all constraints.